

DisorderFlow: Bayesian Flow Networks for Disorder-Aware Antibody CDR-H3 Design

Team 28 — Analyzing Disorder

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iBEC 2026 — International Bioinformatics Engineering Competition
August 2026

1. Executive Summary

DisorderFlow is a computational platform for antibody CDR-H3 sequence design targeting intrinsically disordered protein (IDP) epitopes. The platform combines Bayesian Flow Networks (BFN) with epitope-conditioned likelihood scoring to generate and rank CDR-H3 candidates conditioned on both backbone geometry and epitope disorder profiles.

The project addresses a critical gap in computational antibody design: most existing methods assume rigid, well-ordered epitopes, yet many therapeutically relevant targets (amyloid-beta, tau, alpha-synuclein) are intrinsically disordered. DisorderFlow introduces a disorder-aware scoring framework that couples antigen residue disorder values to CDR-antigen pair features through position-sensitive routing, enabling the model to learn that different regions of a disordered epitope impose different constraints on CDR-H3 sequence selection.

Key achievements include: (1) a rigorously validated epitope-conditioned likelihood shift (ECLS) metric showing positive native-versus-shuffle discrimination across 46 antigen clusters (mean advantage 0.217, 80.4% positive clusters); (2) a deterministic torsion perturbation recovery benchmark (T2.1 v2) achieving 96.2% valid structure fraction with 100% positive contact recovery; and (3) prospective IDP antibody design validation on the 4HIX scaffold where 20 of 20 designed CDR-H3 sequences passed AlphaFold 2 multimer validation, with the top design exceeding native interface quality (ipTM 0.462 vs. native 0.449).

2. Background and Motivation

2.1 The IDP Antibody Design Challenge

Intrinsically disordered proteins (IDPs) lack a stable three-dimensional structure under physiological conditions yet play critical roles in cellular signaling, regulation, and disease. Many neurodegenerative disease targets, including amyloid-beta (Alzheimer's disease), tau, and alpha-synuclein (Parkinson's disease), are IDPs or contain disordered regions.

Antibody-based therapeutics targeting IDP epitopes face a fundamental challenge: the epitope can adopt multiple conformations, and a successful CDR-H3 must accommodate this conformational heterogeneity. Traditional computational antibody design pipelines assume a single, rigid epitope conformation, which is inadequate for disordered targets.

2.2 Bayesian Flow Networks for Sequence Design

Bayesian Flow Networks (BFNs) provide a generative framework for protein sequence design that operates by iteratively refining a probability distribution over amino acid sequences. Unlike autoregressive or diffusion-based approaches, BFNs maintain a discrete categorical distribution at each position and update it through a receiver network that conditions on backbone geometry and structural context.

DisorderFlow extends the BFN framework by introducing a disorder head that predicts per-residue disorder probabilities and a position-sensitive routing mechanism that couples these predictions to the CDR-antigen interaction features.

2.3 Epitope-Conditioned Likelihood Shift (ECLS)

The ECLS metric measures whether a CDR-H3 sequence is preferentially supported by the presence of peptide epitope coordinates. It is defined as:

$$\text{ECLS}(s) = \text{NLL}_{\text{complex}}(s) - \text{NLL}_{\text{peptide-stripped}}(s)$$

where NLL is the mean per-residue negative log-likelihood under ProteinMPNN. A positive ECLS advantage (mean shuffle ECLS minus native ECLS) indicates that the native sequence benefits more from the epitope context than composition-matched controls.

3. Technical Approach

3.1 Architecture Overview

DisorderFlow consists of four major components: (1) BFN Core Module implementing the Bayesian flow network with categorical distributions over 20 amino acids; (2) a Disorder Head predicting per-residue disorder probabilities trained with focal weighting ($\gamma=4.0$); (3) Position-Sensitive Routing coupling antigen residue disorder values to CDR-antigen pair features; and (4) Contrastive Rank Losses teaching the model that factual disorder profiles produce lower NLL than shuffled or mismatched profiles.

3.2 Training Pipeline

The training pipeline uses the SAbDab2 structural database with homology-disjoint splitting (6-axis: PDB identity, paired antibody, VH, VL, H3, antigen). Each training batch contains backbone coordinates, per-residue disorder labels, negative training profiles (shuffled and mismatched), focal-weighted disorder loss, and contrastive rank losses.

3.3 IDP Antibody Design Pipeline

The design pipeline for IDP targets follows five stages: (1) Target Analysis — identify the IDP target and extract epitope region; (2) Disorder Profiling — predict per-residue disorder probabilities; (3) Scaffold Selection — select an appropriate antibody scaffold; (4) CDR-H3 Design — use ProteinMPNN ($T=0.5$, 20 samples) to generate candidates; (5) AF2 Validation — validate each design using AlphaFold 2 multimer with antibody chains passed using colon separator (VH:VL).

4. Results

4.1 ECLS Detection and Temporal Transfer

Adaptation set (46 antigen clusters):

Metric	Value
Mean ECLS advantage	0.217
Median ECLS advantage	0.203
95% Bootstrap CI	[0.146, 0.290]
Positive clusters	80.4%

Temporal final (31 structures, 15 clusters, all post-2021):

Metric	Value
Mean ECLS advantage	0.172
Median ECLS advantage	0.119

95% Bootstrap CI	[0.059, 0.292]
Positive clusters	80.0%

Both frozen gate sets passed, establishing that deposited CDR-H3 sequences carry measurable epitope-conformation-specific signal that transfers to independently deposited structures.

4.2 T2.1 v2 Deterministic Torsion Recovery

The T2.1 v2 benchmark applies deterministic phi/psi torsion perturbation to reach the 1.5-3.0 Å peptide RMSD tier, then tests whether native residue contacts can guide recovery.

Metric	Value
Total structures	31
Torsion hits (1.5-3.0 Å)	26/31 (83.9%)
Valid structures after recovery	25/26 (96.2%)
Mean held-out contact recovery	0.581
95% CI	[0.533, 0.635]
Positive recovery fraction	100%
Gate pass (≥ 0.70)	Yes

This represents a decisive improvement over the initial T2 protocol (3/7 valid, -0.017 recovery), demonstrating that the previous failure was caused by parameter calibration rather than a fundamental limitation.

4.3 4HIX IDP Design Validation

Using the 4HIX scaffold (humanized 3D6 Fab, Abeta 1-6 epitope DAEFRH), ProteinMPNN generated 20 CDR-H3 candidates. All 20 passed AF2 multimer validation.

Rank	H3 Sequence	ipTM	pLDDT	iPAE (Å)
Native	VRYDHYSGSSDY	0.449	0.194	24.7
1	LYDESKDAESE	0.462	0.200	24.5
2	LYDAHGAHSL	0.461	0.192	24.4
3	LYDGSIGAESQ	0.460	0.195	24.5
4	LYDSSVDASGH	0.459	0.199	24.7
5	LFNEANCAESY	0.458	0.199	24.3

The top 3 designs exceed native ipTM, demonstrating that the BFN-derived framework can guide CDR-H3 design with computational interface quality equal to or exceeding the deposited native sequence.

4.4 Critical Methodological Finding: AF2 Chain Separator

During AF2 validation, we discovered that antibody chains must be passed with a colon separator (VH:VL) rather than concatenated (VH+VL). Without the chain break, ipTM dropped from ~0.45 to ~0.10, a 4.5-fold underestimation of interface quality. This fix was incorporated into the design pipeline and is essential for accurate AF2 evaluation of antibody-antigen complexes.

5. Evidence Hierarchy

From strongest to weakest: (1) ECLS temporal final (n=15 clusters, post-2021, sealed) — positive native-versus-shuffle discrimination on independent structures; (2) ECLS adaptation (n=46 clusters) — replicates the temporal final pattern on hold-out data; (3) T2.1 v2 torsion recovery (n=31 structures, 96.2% valid) — physical recovery of peptide contacts after deterministic perturbation; (4) 4HIX design validation (n=20 designs, 20/20 AF2 pass) — prospective design with top ipTM exceeding native; (5) T1 ensemble (n=7 clusters) — establishes single-pose overconfidence; (6) Generator calibration (n=7 clusters) — exploratory; calibrated reranker works, universal reranker rejected.

6. Software Implementation

The codebase is organized into modular components: the BFN core module with categorical distributions and focal-weighted losses; the disorder-augmented dataset with label injection and contrastive negative profiles; the receiver network with position-sensitive routing; the IDP antibody design pipeline with five stages; and the AF2 validation infrastructure with proper chain separator handling. All code is provided with frozen YAML configuration contracts and SHA256 manifests for reproducibility.

7. Reproducibility

All results are reproducible from frozen configs and checksummed data. T2.1 v2 per-structure results are provided in `results_t2.1_v2_final.json` (31 per-structure records with torsion traces). 4HIX design results are provided in `idp_design_results/4hix_final_validation/final_report.json` (20 designs with AF2 metrics). An integrative analysis document (`INTEGRATIVE_ANALYSIS.md`) summarizes the complete evidence chain. SHA256 manifests are provided for all result files.

8. Limitations

(1) No wet-lab validation: all results are computational. No experimental binding, affinity, or specificity data are available. (2) Short epitope: the 4HIX validation uses a 6-residue epitope (DAEFRH); AF2 interface metrics are less discriminative for very short peptides. (3) Training set overlap: the 4HIX Fab was deposited in 2012 and was in AF2's training set. (4) Disorder head evaluation: the retrained disorder head has not yet been evaluated on the CAID benchmark. (5) Single IDP target: only 4HIX has been validated; generalization to other IDP targets remains to be demonstrated. (6) Composition-matched shuffles are counterfactual controls, not experimental non-binders.

9. Future Work

Planned extensions include: (1) CAID benchmark evaluation of the retrained disorder head; (2) Abeta redesign with the fixed model; (3) Extended IDP targets (tau, alpha-synuclein); (4) Experimental validation by SPR, ELISA, or biolayer interferometry; (5) Manuscript submission to Bioinformatics journal; (6) 4HIX T1 ensemble MD simulation to characterize conformational heterogeneity.

10. Published Results and Data Availability

No peer-reviewed publications have resulted from this work as of August 2026. A manuscript draft is in preparation for submission to Bioinformatics. All code, frozen configs, result files, and reproducibility documentation are available in the DisorderFlow repository. Third-party datasets and model weights are distributed separately by checksum.

11. Team and Contributions

Member	Role	Contribution
Chen Haoyang	Team Lead	Project design, BFN architecture, ECLS methodology, manuscript
Hu Jing	Comp. Biology	IDP pipeline, AF2 validation, disorder head training
Yu Jiarui	Software Eng.	Training infrastructure, benchmarks, reproducibility

12. References

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